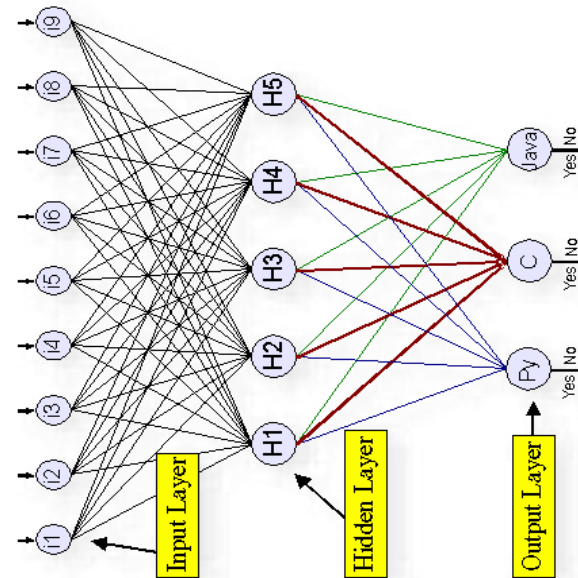




AI Current Development, Neural Networks and Deep Learning

Ioannis Tsioulis (MC)

Outline

1. AI recent development
2. ANN
3. Deep Learning

AI Recent Development (1)

Deep Learning:

- Deep learning is a subset of AI that uses artificial neural networks to simulate the learning process of the human brain.
- Deep learning has been used to develop models for image and speech recognition, natural language processing, and other applications.
- Some of the most significant recent developments in deep learning include the development of convolutional neural networks (CNNs) for image recognition and recurrent neural networks (RNNs) for natural language processing.

AI Recent Development (2)

Combining with Reinforcement Learning:

- Reinforcement learning is a type of AI that focuses on training agents to make decisions in an environment in order to maximize a reward signal.
- Reinforcement learning has been used to develop models for game playing, robotics, and other applications where agents need to make decisions in complex environments.
- Some of the most significant recent developments in reinforcement learning include the development of deep reinforcement learning algorithms, such as the Deep Q-Network (DQN), which uses deep neural networks to estimate Q-values for state-action pairs.

AI Recent Development (3)

Generative Language Models:

- Generative language models are a type of AI model that can generate human-like responses to natural language prompts.
- The first generative language models were based on statistical methods, but they were limited in their ability to produce coherent and meaningful responses.
- In recent years, deep learning has been used to develop more advanced generative language models, such as OpenAI's GPT (Generative Pre-trained Transformer) series.

AI Recent Development (4)

The Transformer:

- A type of neural network architecture that can be used for a variety of machine learning applications, including natural language processing (NLP), computer vision, and speech recognition
- Processes entire sequences of data in parallel, making it more computationally efficient than traditional sequential models
- Uses self-attention mechanisms to focus on different parts of the input sequence depending on the context, improving performance on a variety of tasks

AI Recent Development (5)

ChatGPT4:

- ChatGPT4 is the latest version of OpenAI's generative language model that uses deep learning to generate human-like responses to natural language prompts.
- ChatGPT4 is based on the transformer architecture, which allows it to process large amounts of text and generate responses that are both coherent and contextually appropriate.
- ChatGPT4 has over 10 trillion parameters, making it one of the largest language models in existence.

AI Recent Development (6)

Multi-modality Models:

- A type of AI model that can process and integrate information from multiple sources, such as text, images, and audio
- Designed to handle complex data that is represented in multiple modalities
- Use deep learning techniques such as CNNs and RNNs to extract features from each modality, which are then combined and processed by the model
- Driven by the increasing availability of data that includes multiple modalities and the need for AI systems to process complex real-world scenarios

ChatGPT development framework

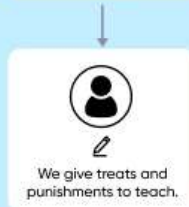
Step 1 →

Collect demonstration data and train a supervised policy.

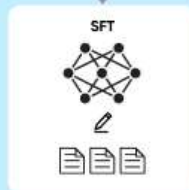
A prompt is sampled from our prompt dataset.



A labeler demonstrates the desired output behavior.



This data is used to fine-tune GPT-3.5 with supervised learning

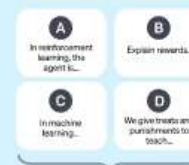


Source: openai.com

Step 2 →

Collect comparison data and train a reward model.

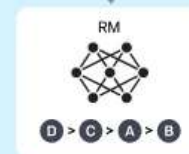
A prompt and several model outputs are sampled



A labeler ranks the outputs from best to worst.



This data is used to train our reward model.



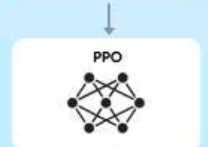
Step 3 →

Optimize a policy against the reward model using the PPO reinforcement learning algorithm

A new prompt is sampled from the dataset.



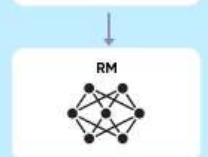
The PPO model is initialized from the supervised policy.



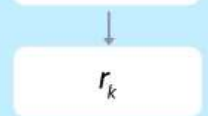
The policy generates an output.



The reward model calculates a reward for the output.

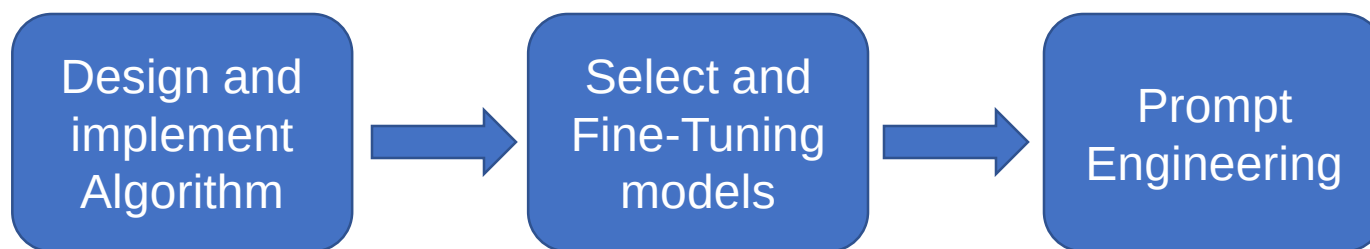


The reward is used to update the policy using PPO.



net solutions

Trend





ANN History

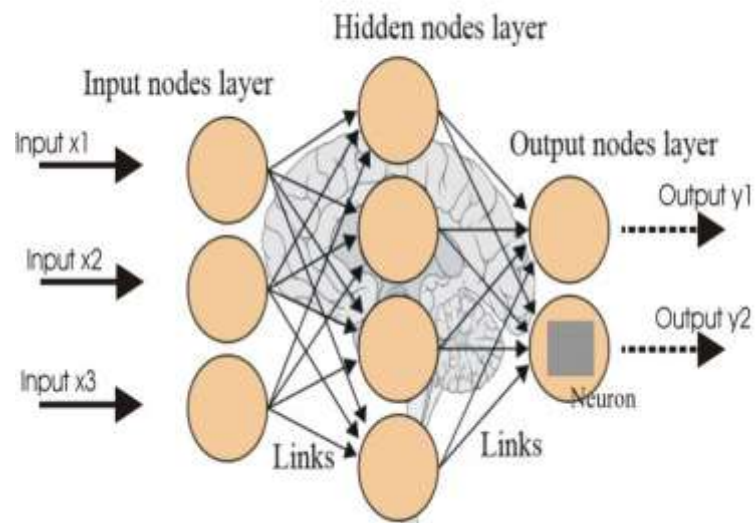
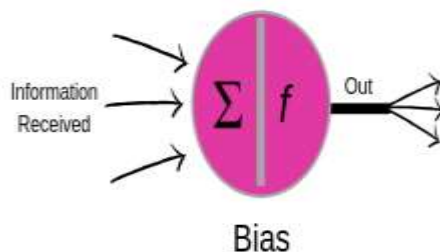
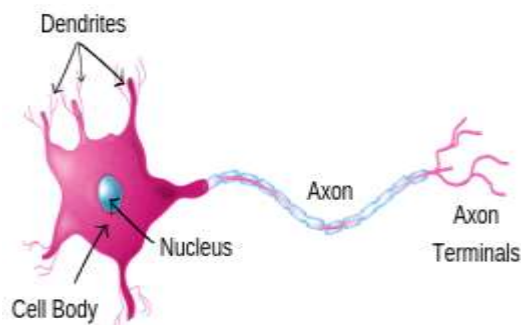
ANN isn't a new idea, it has been introduced far back to 1943 by Warren McCulloch and Walter Pitts' research paper ["A Logical Calculus of the Ideas Immanent in Nervous Activity"](#). The paper describes a network of artificial neurons model that can solve some logic problems.

In 1969, when a formal self-justification mathematic algorithm called backpropagation is introduced, then ANN is widely known and used.



Motivations of ANN

Take inspiration from human brain



←backpropagation

A close-up photograph of a ginger and white cat looking directly at the camera. The cat has bright yellow-green eyes and a white patch on its face and chest. The background is blurred, showing warm tones.

imread



3-channel matrix

Blue

Green

Red

		255	134	93	22
		255	134	202	22
	255	231	42	22	4
255	123	94	83	2	92
34	44	187	92	34	

im2vector
(or flatten)



reshaped image vector

$$\begin{pmatrix} 255 \\ 231 \\ 42 \\ 22 \\ 123 \\ 94 \\ \vdots \\ \vdots \\ 92 \\ 142 \end{pmatrix}$$

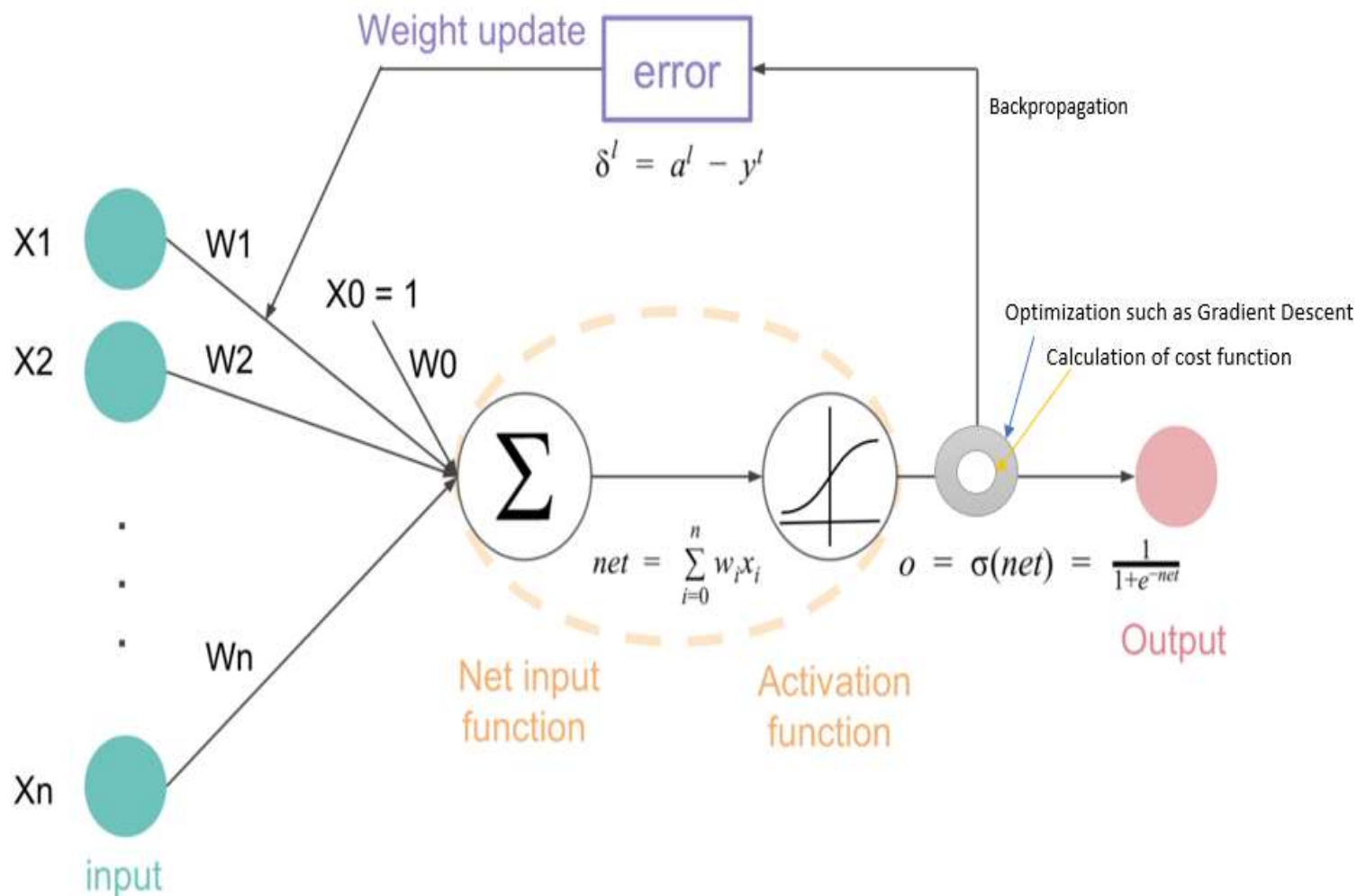


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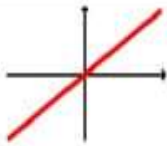
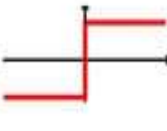
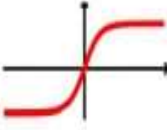
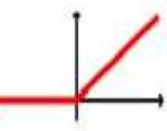
$\begin{pmatrix} 255 \\ 231 \\ 42 \\ 22 \\ 123 \\ 94 \\ \vdots \\ \vdots \\ 92 \\ 142 \end{pmatrix}$

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Sensitivity:
Internal

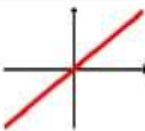
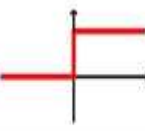
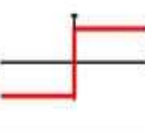


Activation Functions

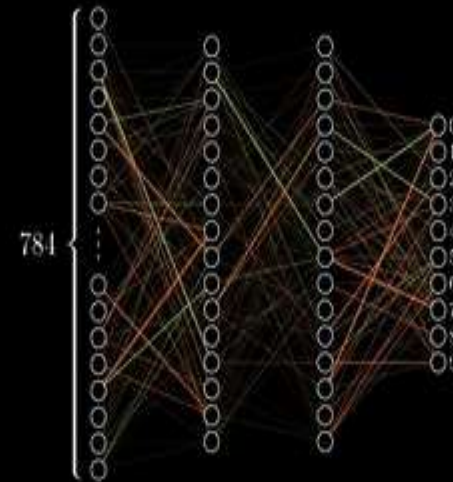
Activation Function	Equation	Example	1D Graph
Linear	$\phi(z) = z$	Adaline, linear regression	
Unit Step (Heaviside Function)	$\phi(z) = \begin{cases} 0 & z < 0 \\ 0.5 & z = 0 \\ 1 & z > 0 \end{cases}$	Perceptron variant	
Sign (signum)	$\phi(z) = \begin{cases} -1 & z < 0 \\ 0 & z = 0 \\ 1 & z > 0 \end{cases}$	Perceptron variant	
Piece-wise Linear	$\phi(z) = \begin{cases} 0 & z \leq -1/2 \\ z + 1/2 & -1/2 \leq z \leq 1/2 \\ 1 & z \geq 1/2 \end{cases}$	Support vector machine	
Logistic (sigmoid)	$\phi(z) = \frac{1}{1 + e^{-z}}$	Logistic regression, Multilayer NN	
Hyperbolic Tangent (tanh)	$\phi(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$	Multilayer NN, RNNs	
ReLU	$\phi(z) = \begin{cases} 0 & z < 0 \\ z & z > 0 \end{cases}$	Multilayer NN, CNNs	



Activation Functions

Activation Function	Equation	Example	1D Graph
Linear	$\phi(z) = z$	Adaline, linear regression	
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Sign (signum)	$\phi(z) = \begin{cases} -1 & z < 0 \\ 0 & z = 0 \\ 1 & z > 0 \end{cases}$	Perceptron variant	
Piece-wise Linear	$\phi(z) = \begin{cases} 0 & z \leq -\frac{1}{2} \\ z + \frac{1}{2} & -\frac{1}{2} \leq z \leq \frac{1}{2} \\ 1 & z \geq \frac{1}{2} \end{cases}$	Support vector machine	
Logistic (sigmoid)	$\phi(z) = \frac{1}{1 + e^{-z}}$	Logistic regression, Multilayer NN	
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ReLU	$\phi(z) = \begin{cases} 0 & z < 0 \\ z & z > 0 \end{cases}$	Multilayer NN, CNNs	

Training in progress...



Sensitivity:
Internal

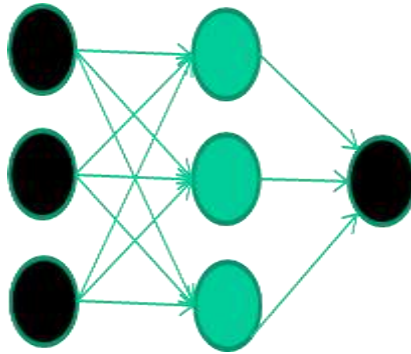
So, what is Deep Learning?

The short answers

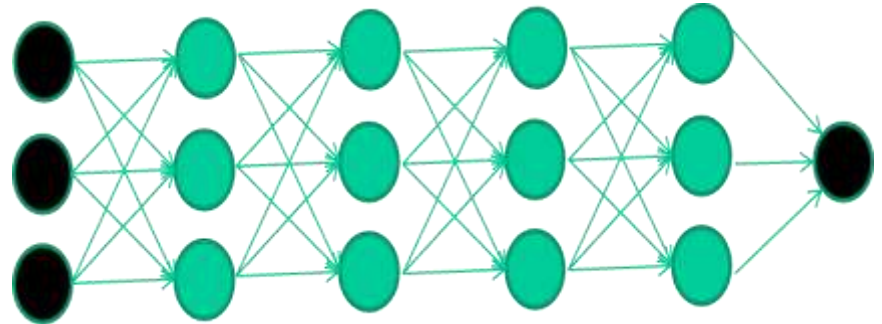
‘Deep Learning’ means using a neural network with many layers of nodes (there is no accurate numbers here) between input and output

The series of layers between input & output do feature identification and processing in a series of stages

1 vs Deep



1-hidden layer



Deep-hidden layers

Again, we need to think of trade off balance between computation complexity and performance

Deep learning algorithms

There are many different deep learning algorithm designed for dealing with different problems. Some examples are:

- Convolutional Neural Network (CNN)
- Recurrent Neural Networks (RNNs)
- Long Short-Term Memory Networks (LSTMs)
- Stacked Auto-Encoders
- Deep Boltzmann Machine (DBM)
- Deep Belief Networks (DBN)

Computer Vision and Deep Learning

CNN: Convolutional Neural Network

Yann LeCun, Professor of Computer Science
The Courant Institute of Mathematical Sciences
New York University
Room 1220, 715 Broadway, New York, NY 10003, USA.
(212)998-3283 yann@cs.nyu.edu



In 1995, Yann LeCun and Yoshua Bengio introduced the concept of convolutional neural networks.

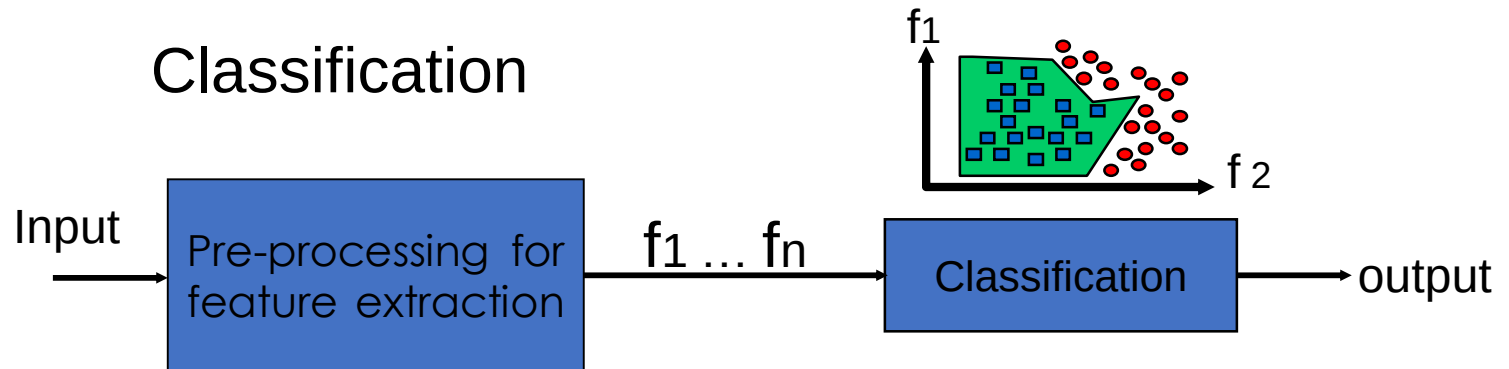
A path towards human-level intelligence
(01/04/2022)

https://www.youtube.com/watch?v=_K0h7oEe8BQ&list=PLZXqDfYHTYo4U_Q-LVK72xzRe2JhlwOzl&index=19

About CNN's

- CNN's Were **neurobiological** motivated by the findings of locally sensitive and implicitly extracts relevant features.
- Convolutional Neural Networks are a special kind of **multi-layer neural networks** with **feed-forward** network that can extract topological properties from an image.
- Like almost every other neural networks they **are trained** with a version of the **back-propagation algorithm**.
- Convolutional Neural Networks are designed to **recognize visual patterns** directly from pixel images with minimal preprocessing.
- They can recognize patterns with extreme variability (such as handwritten characters).

Classification



Convolutional neural network



CNN by Example

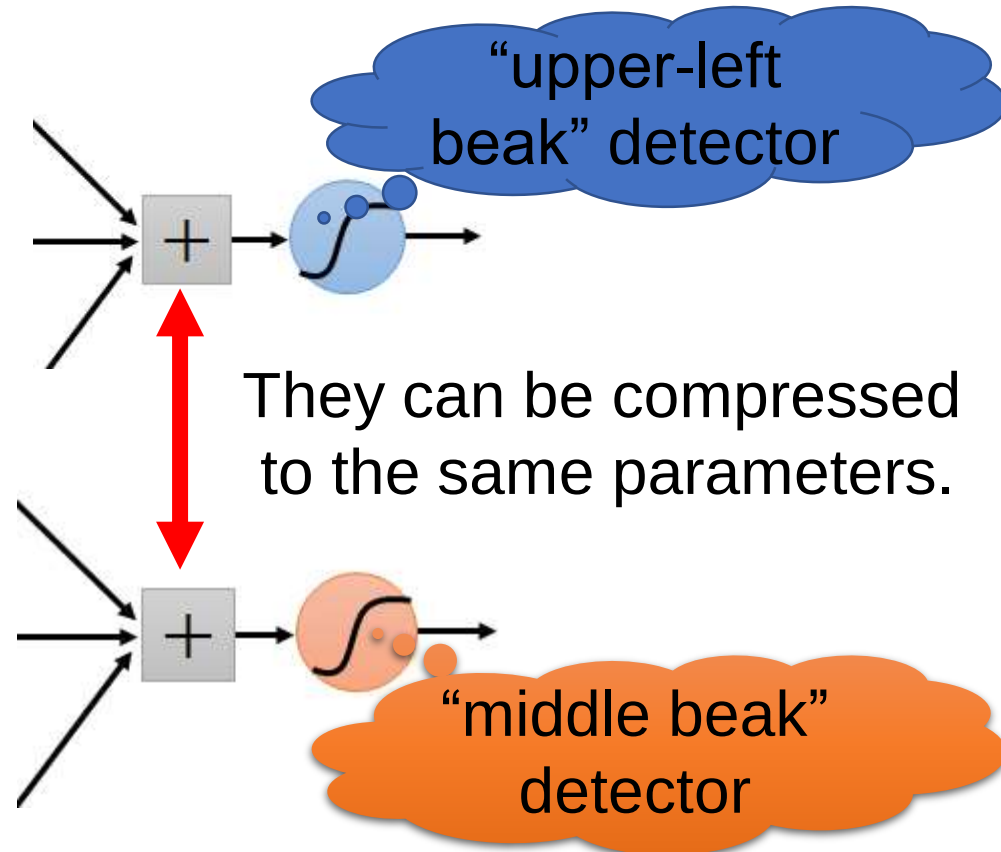
- Some patterns are much smaller than the whole image

Can represent a small region with fewer parameters



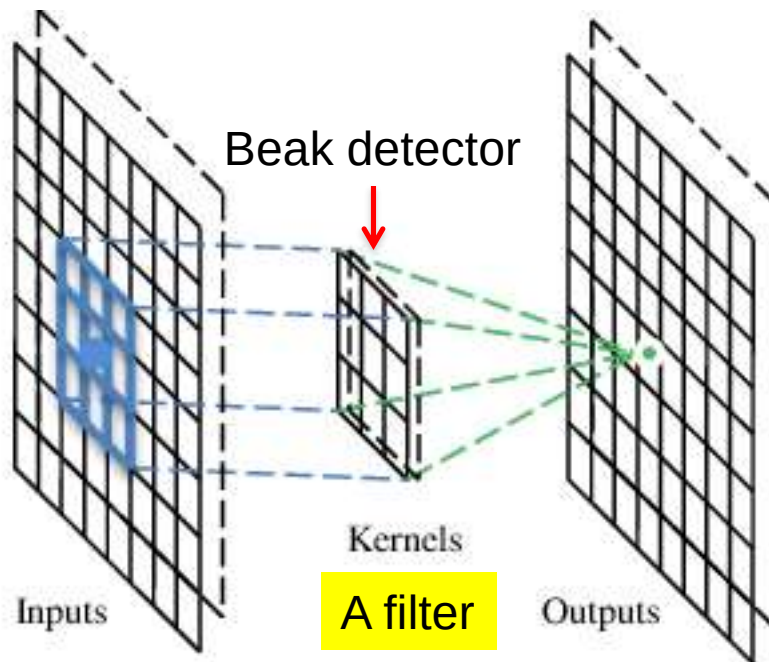
Same pattern appears in different places: They can be compressed!

What about training a lot of such “small” detectors and each detector must “move around”.



A convolutional layer

A CNN is a neural network with some convolutional layers (and some other layers). A convolutional layer has a number of filters that does convolutional operation.



Convolution

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

These are the network parameters to be learned.

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

⋮ ⋮

Each filter detects a small pattern (3 x 3).

Convolution

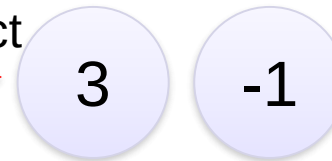
stride=1

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

Dot
product



6 x 6 image

Convolution

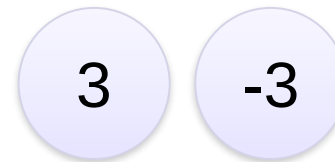
If stride=2

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1



Convolution

stride=1

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

1	-1	-1
-1	1	-1
-1	-1	1

Filter 1

3	-1	-3	-1
-3	1	0	-3
-3	-3	0	1
3	-2	-2	-1

Convolution

stride=1

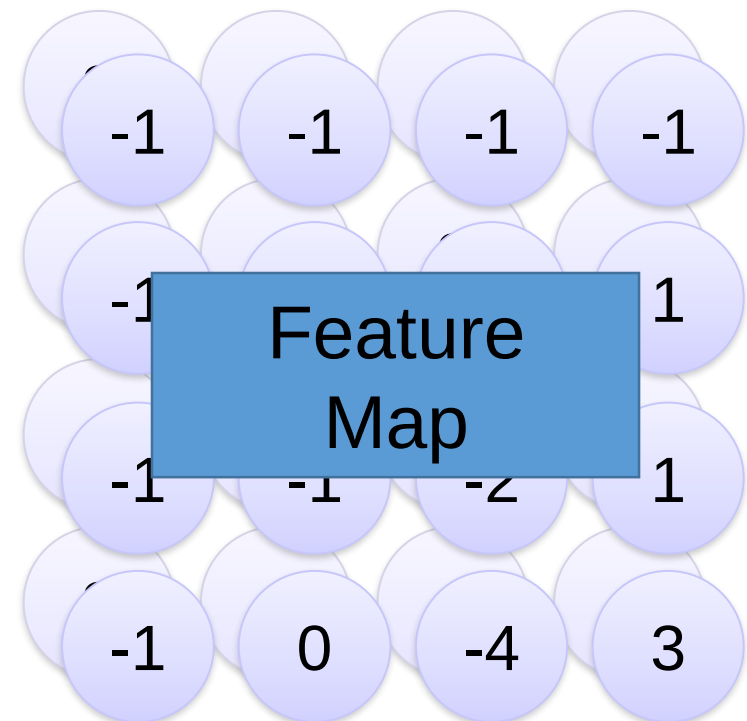
1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

6 x 6 image

-1	1	-1
-1	1	-1
-1	1	-1

Filter 2

Repeat this for each filter



Two 4 x 4 images
Forming 2 x 4 x 4 matrix

Color image: RGB 3 channels

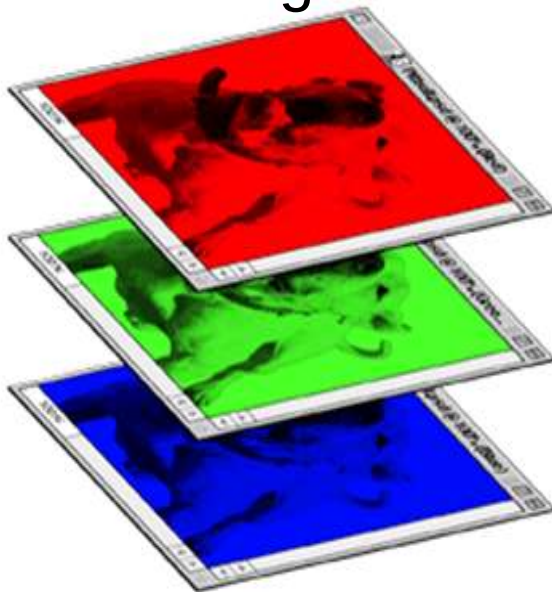
$$\begin{bmatrix} 1 & -1 & -1 \\ -1 & 1 & -1 \\ -1 & -1 & 1 \end{bmatrix}$$

Filter 1

$$\begin{bmatrix} -1 & 1 & -1 \\ -1 & 1 & -1 \\ -1 & 1 & -1 \end{bmatrix}$$

Filter 2

Color image



$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \end{bmatrix}$$



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Convolution v.s. Fully Connected

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0

image

1	-1	-1
-1	1	-1
-1	-1	1

-1	1	-1
-1	1	-1
-1	1	-1

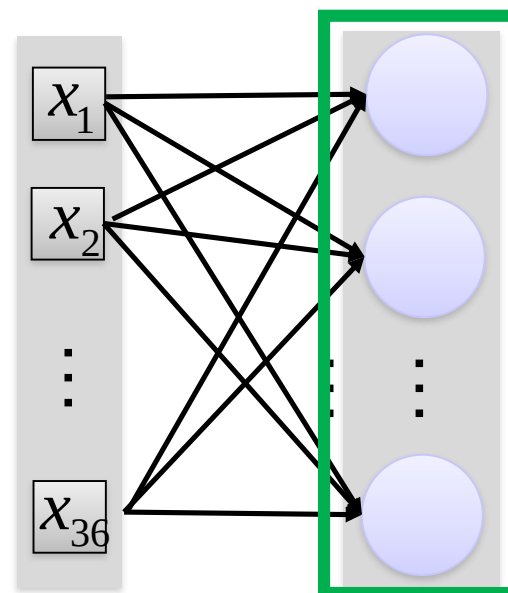


convolution

-1	-1	-1	-1
-1	-1	-2	1
-1	-1	-2	1
-1	0	-4	3

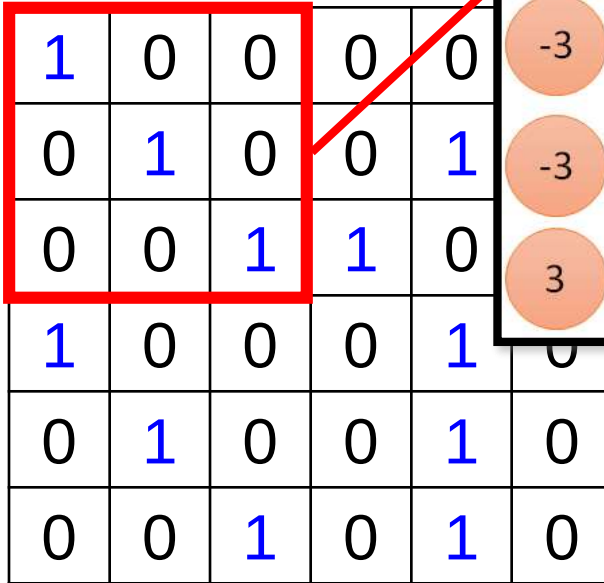
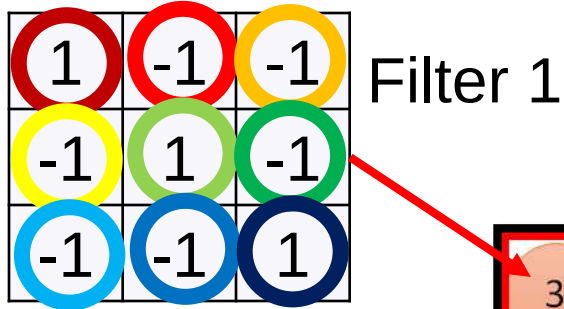
Fully-
connected

1	0	0	0	0	1
0	1	0	0	1	0
0	0	1	1	0	0
1	0	0	0	1	0
0	1	0	0	1	0
0	0	1	0	1	0



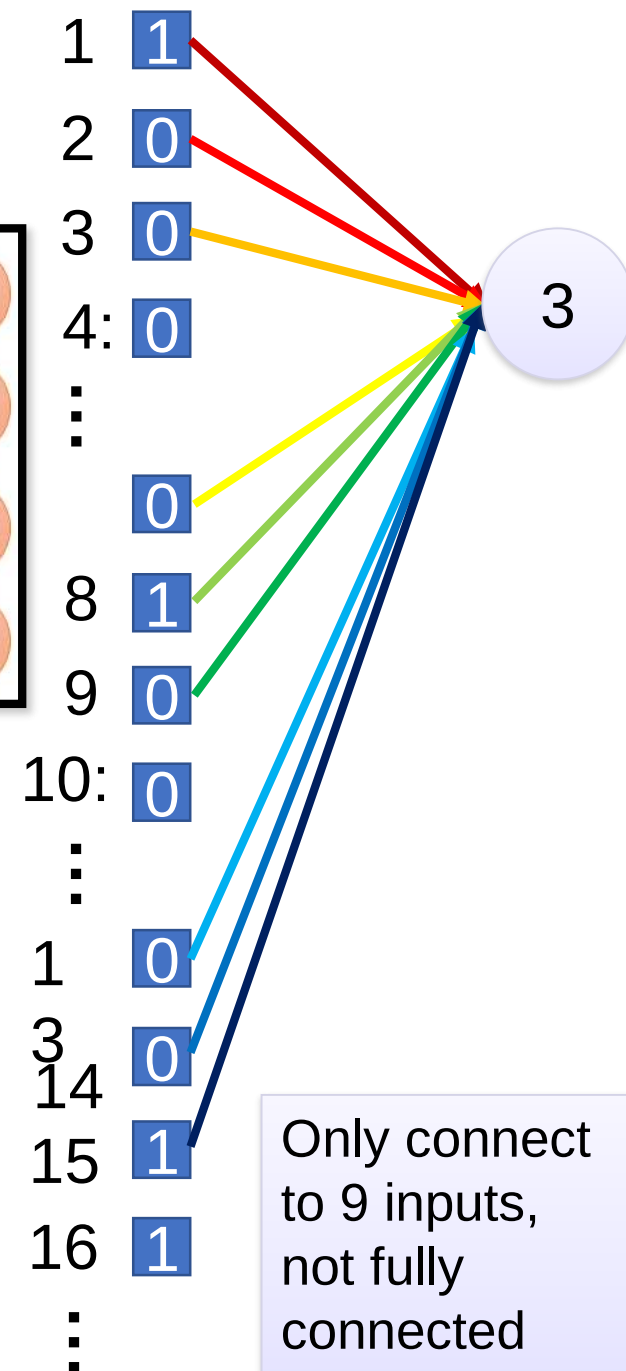
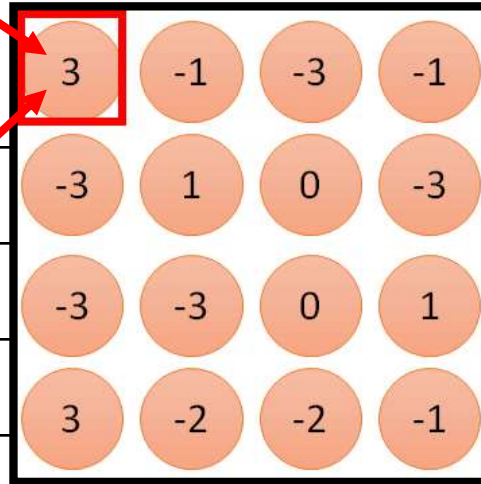
TOP 30
GUARDIAN
UNIVERSITY GUIDE 2019

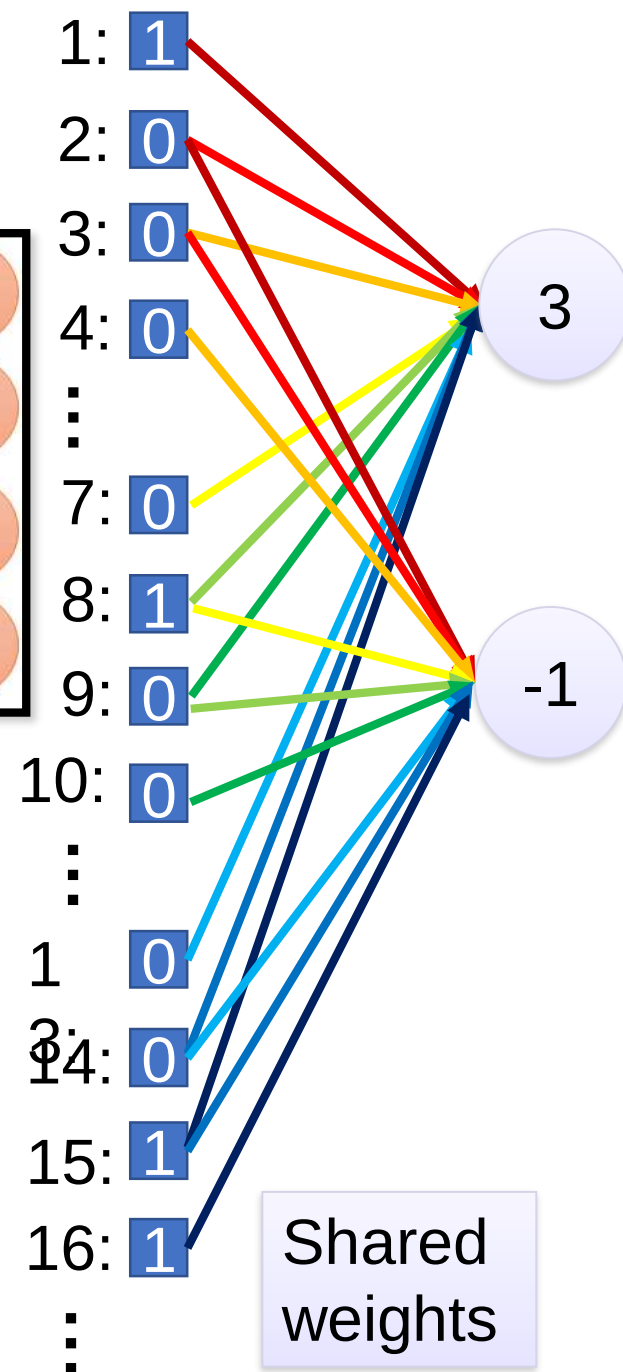
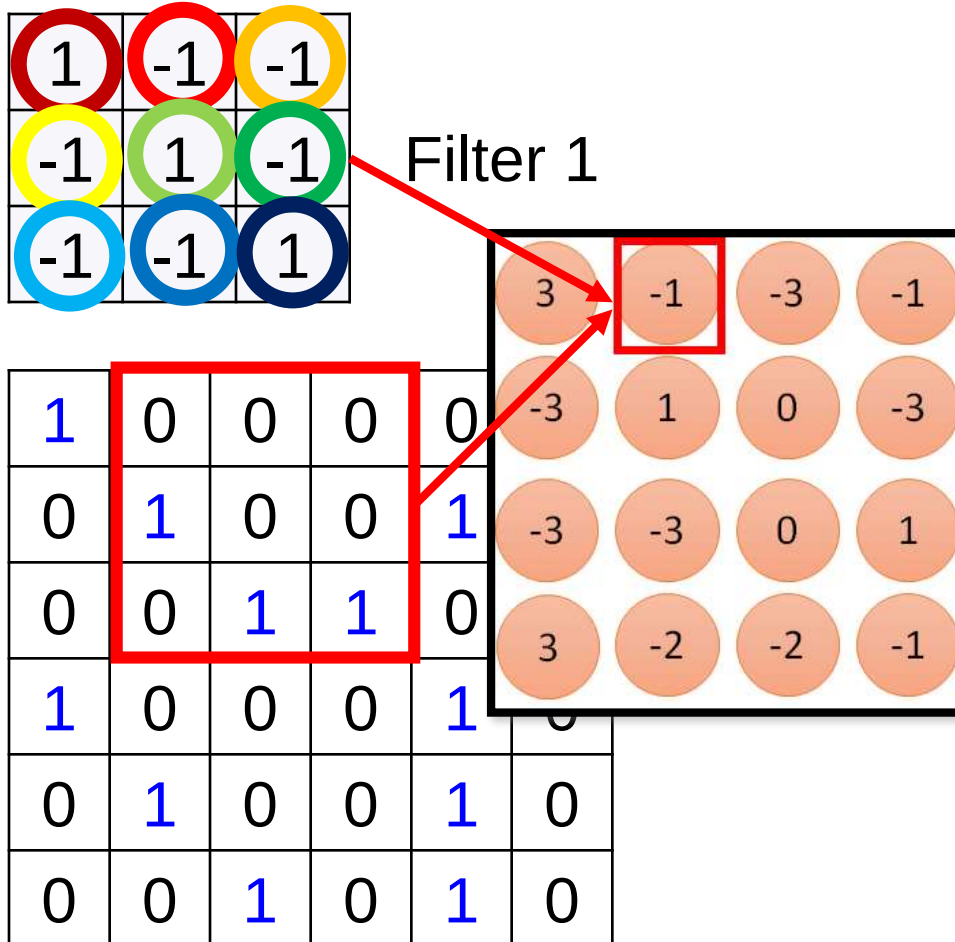
derby.ac.uk



6 x 6 image

fewer parameters!



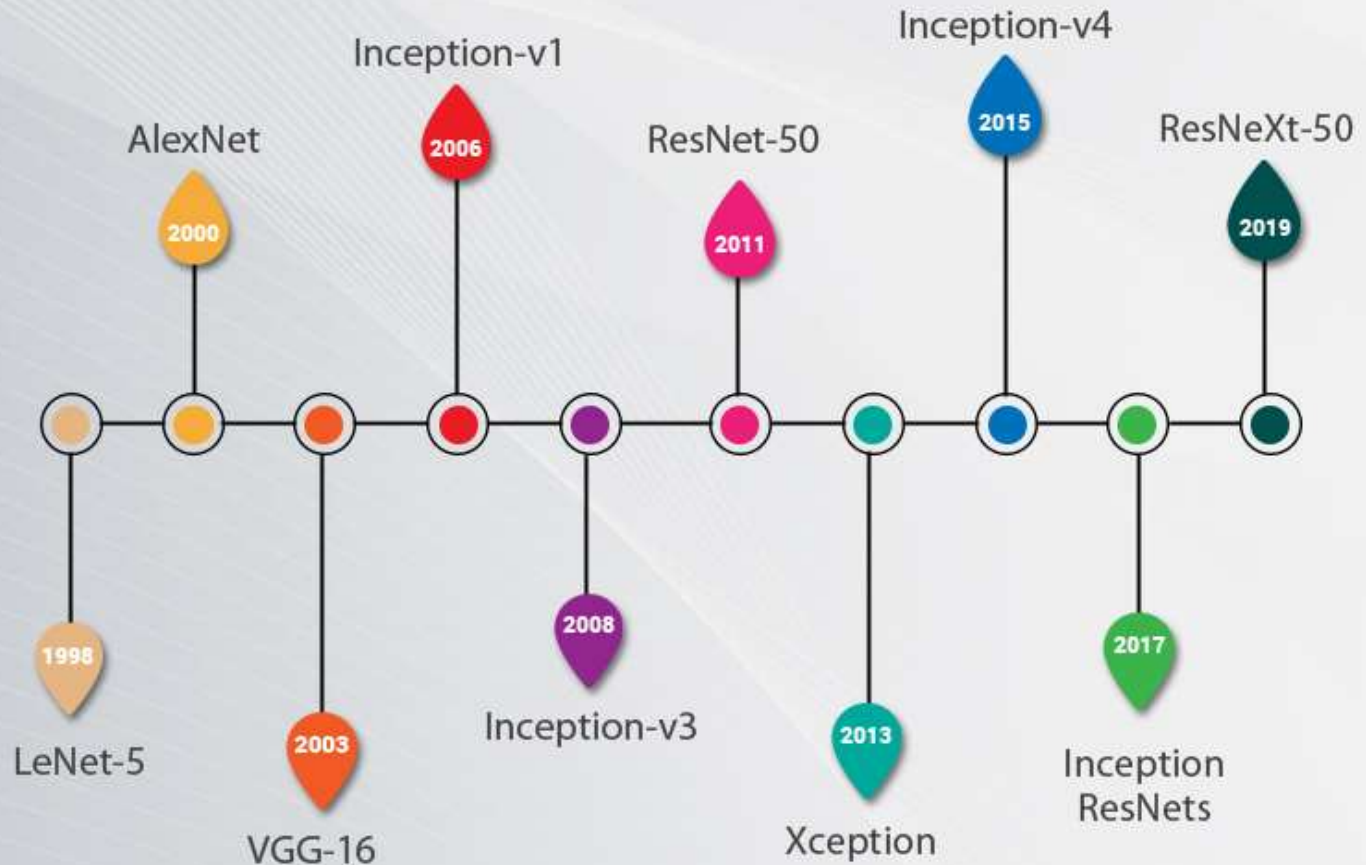


Fewer parameters

Even fewer parameters

Different CNN Architectures

CNN architectures over a timeline(1998-2019)



Differences in examples

LeNet

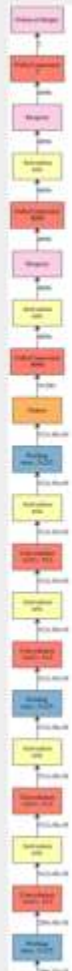
AlexNet

VGG

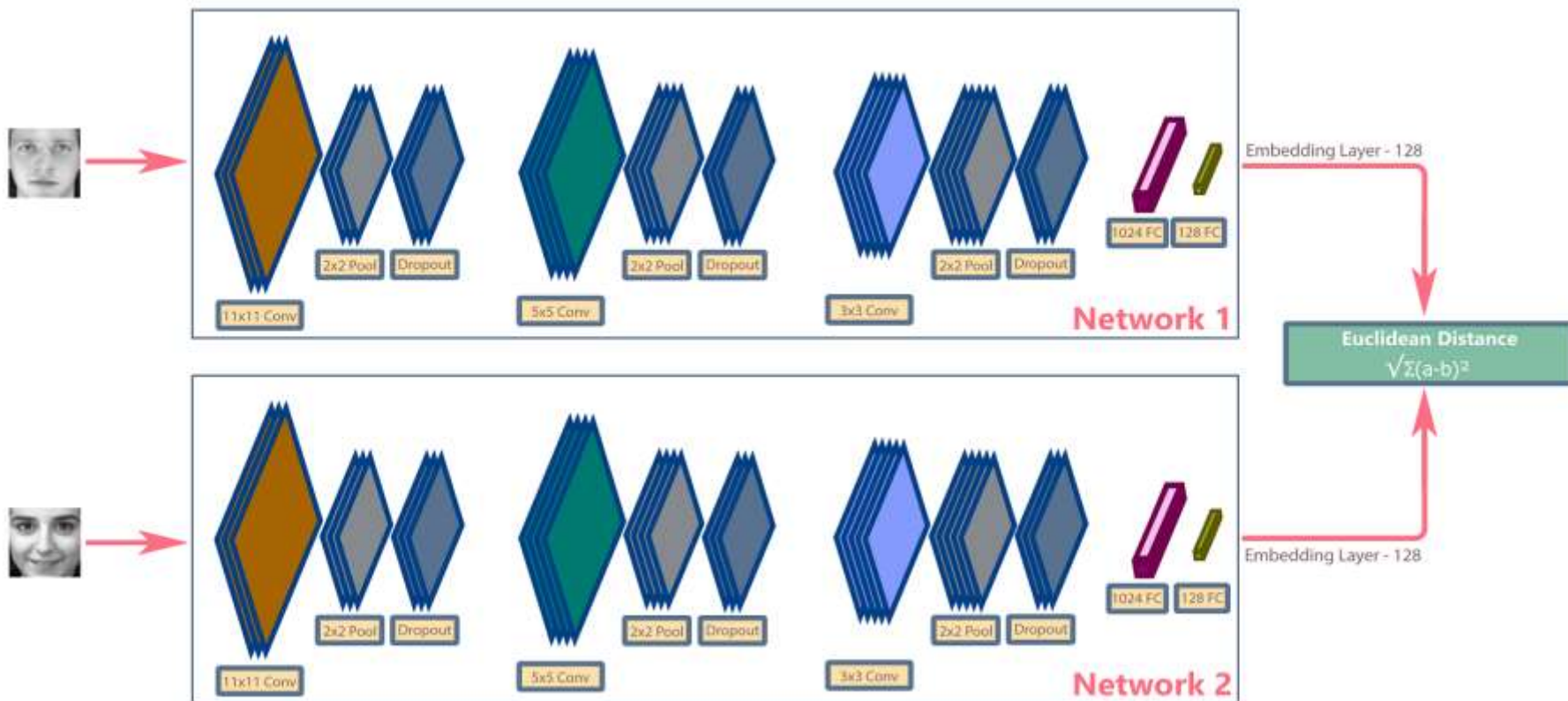
GoogLeNet

Inception V3

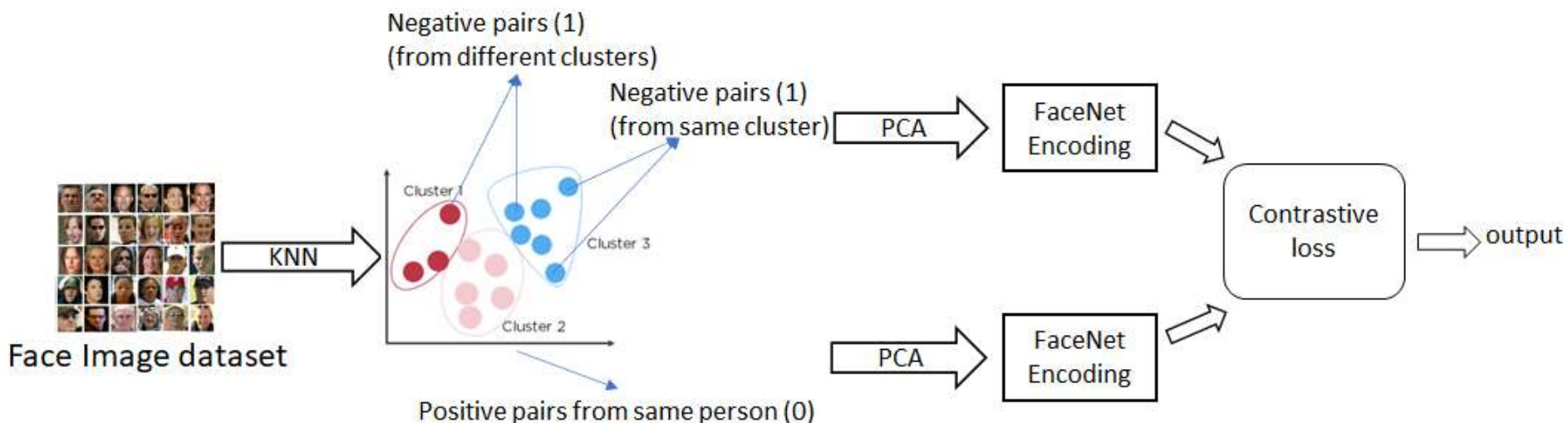
Inception BN



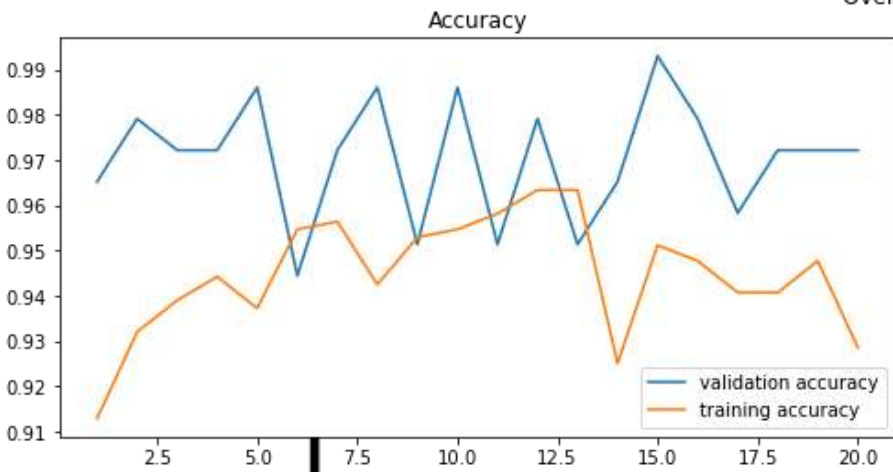
Contrastive Deep Learning (few short) algorithms



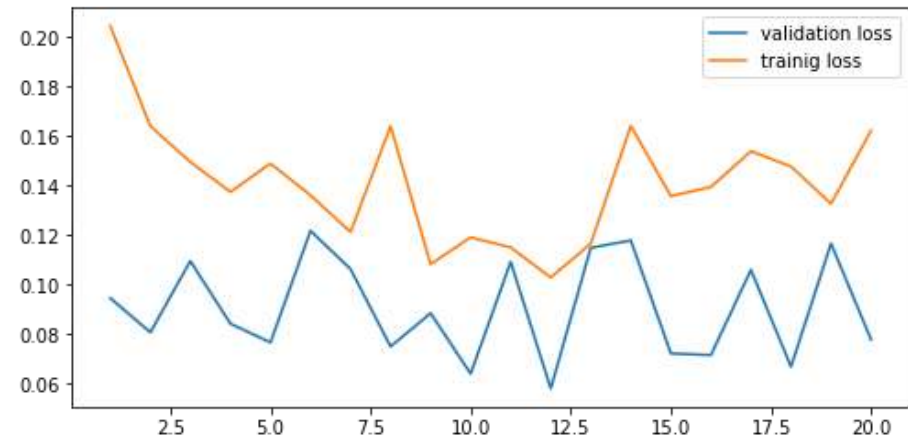
One possible improvement which I am working on



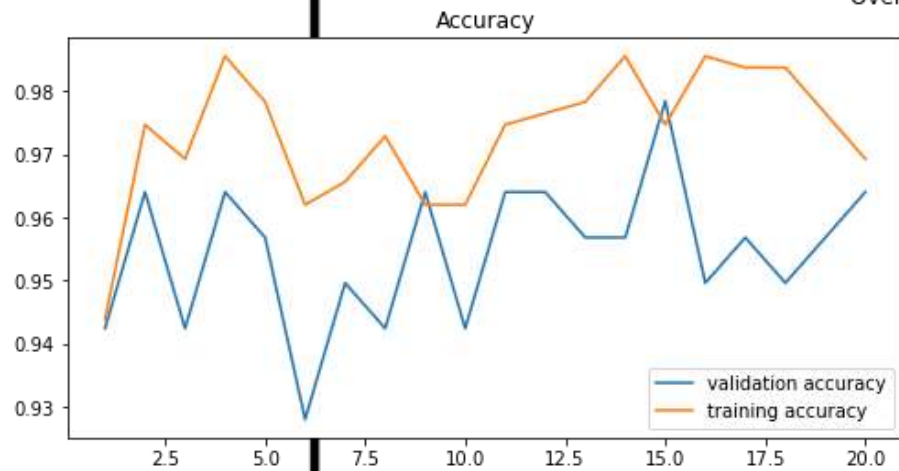
Overfitting analysis



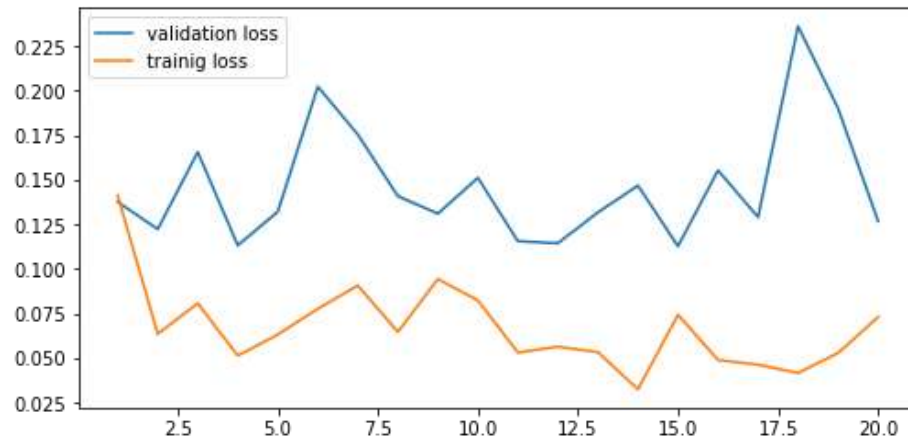
Loss



Overfitting analysis



Loss



TOP 30
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<https://www.kaggle.com/code/harryyu/cluster-siamese-model-on-face-recognition-1-data>

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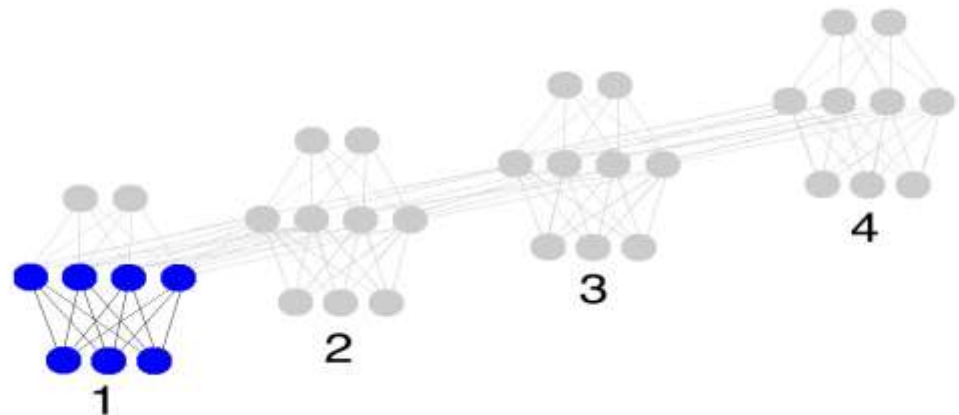
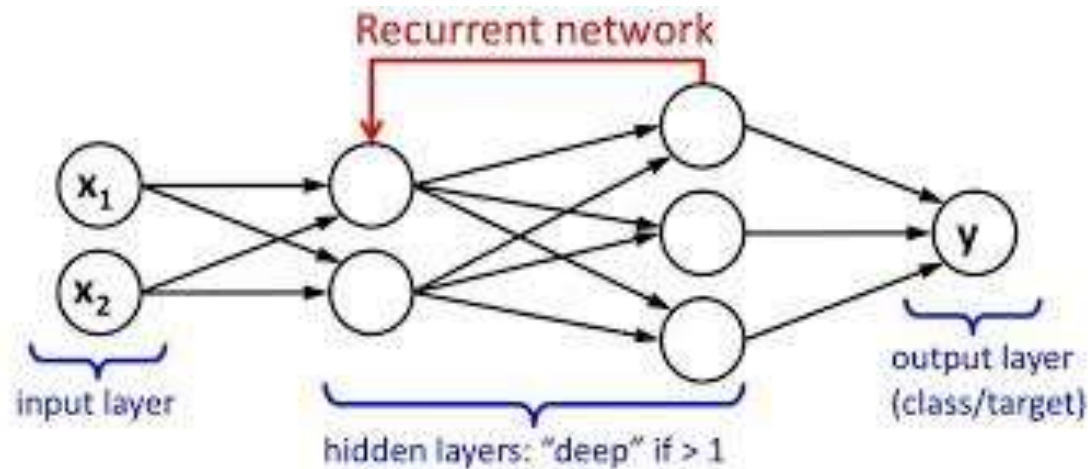
Sensitivity:
Internal

Other types of DNN

We will talk about them in next time lab session

Recurrent Neural Networks

Recurrent networks, take as their input not just the current input example they see, but also what they have perceived previously in time.



The Problem of Long-Term Dependencies

Ex. 1: I used to live in France.

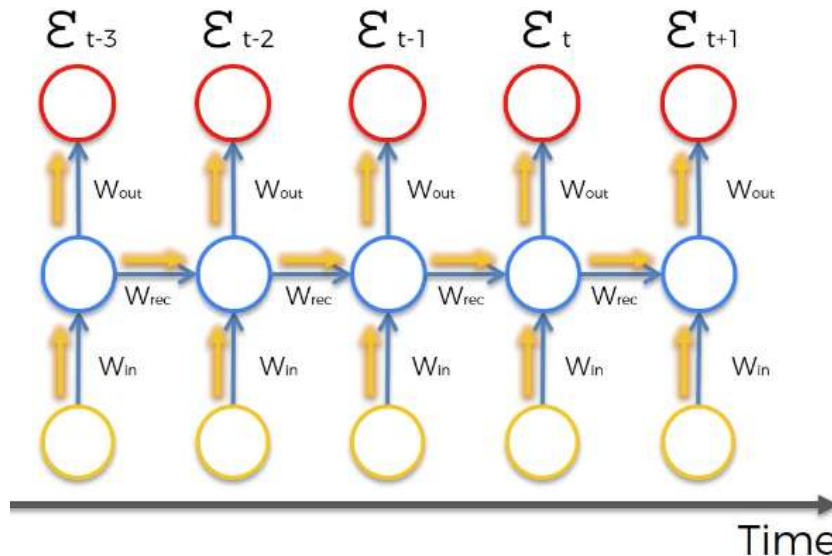
I learnt to speak ?.

RNN is good

Ex. 2: I grew up in France ... (a long sentence)

I speak fluent ?.

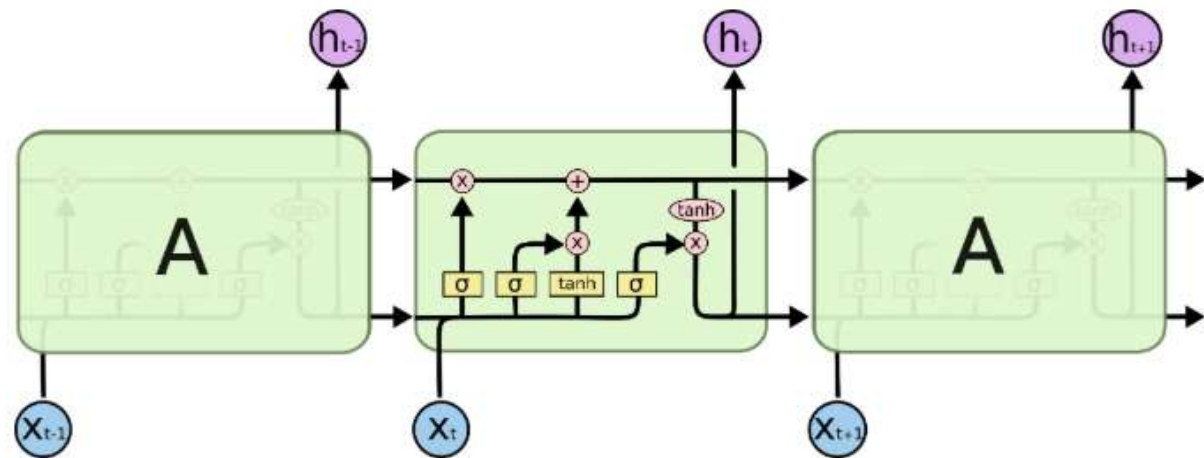
RNN theoretically should be good, but not in practices



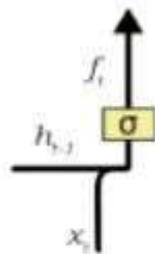
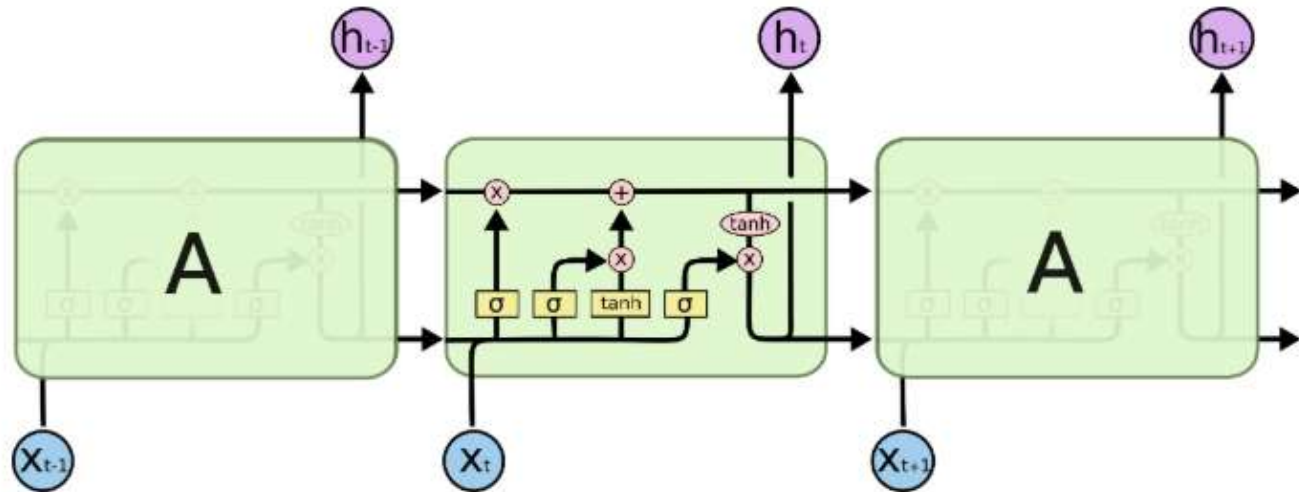
Vanishing gradient problem

LSTM (Long Short Term Memory networks)

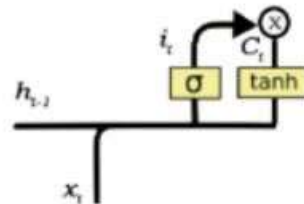
- LSTM is a special kind of recurrent neural network capable of learning long-term dependencies.
- They were introduced by Hochreiter & Schmidhuber in 1997.



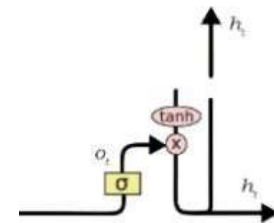
How does LSTM work?



Forget gate



Input gate



Output gate

Pre-trained models

Pre-trained learning models are good sources for solving a common AI problem. Due to time restrictions or computational restraints, it's not always possible to build a model from scratch which is why pre-trained models are useful. In addition, you can use a pre-trained model as a benchmark to either improve the existing model or test your own model against it.

Pre-trained CV models e.g YOLO (Object recognition), CIFAR10 (image classification based on ImageNet)

Pre-trained NLP models e.g. BERT, GPT-3...

Transfer learning process

From a practical perspective, the entire transfer learning process can be summarised as follows:

- 1. Select a pre-trained model based on your task**
- 2. Classify your problem according to the Size-Similarity Matrix**
- 3. Fine-tune your model**

Generative Adversarial Networks (GANs)

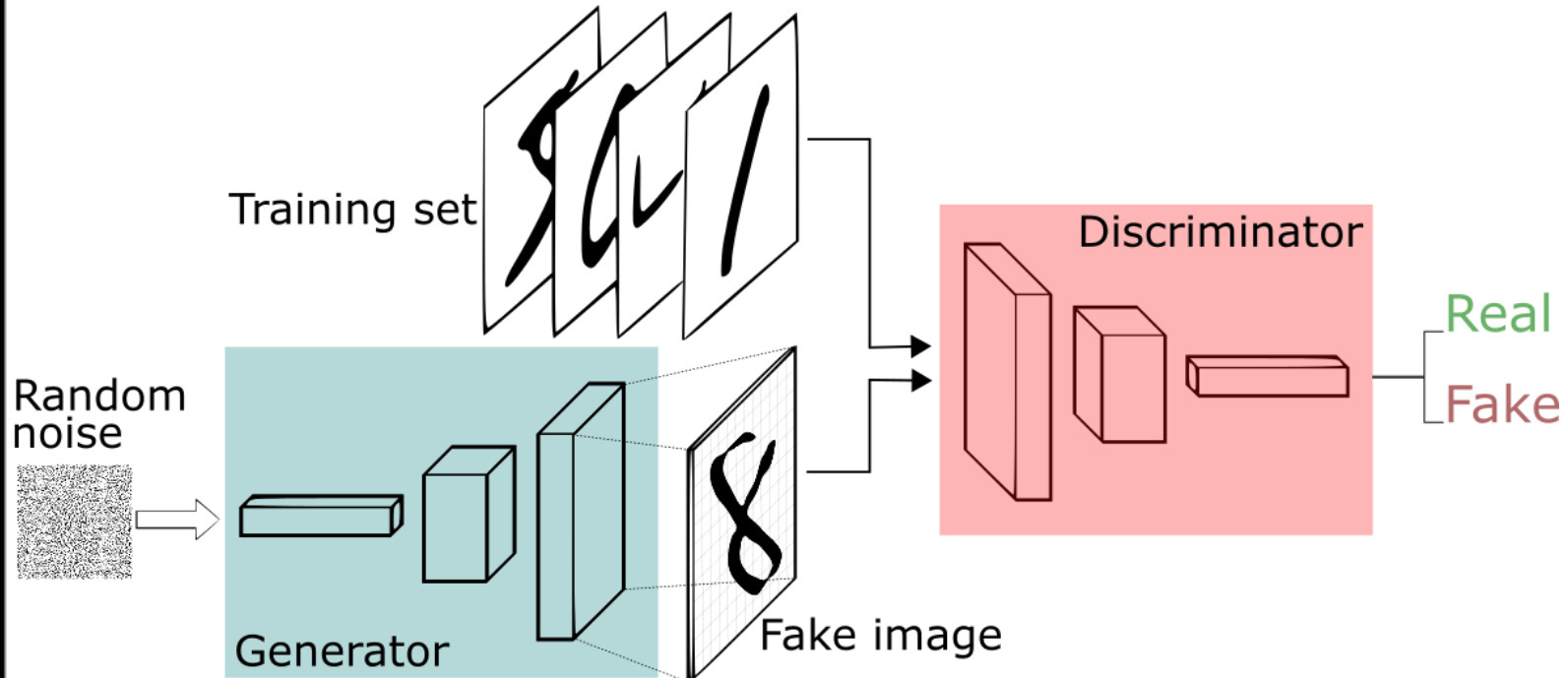
- Generative Adversarial Networks, or GANs for short, are an approach to generative modelling using deep learning methods, such as convolutional neural networks.
- Generative modelling is an unsupervised learning task in machine learning that involves automatically discovering and learning the regularities or patterns in input data in such a way that the model can be used to generate or output new examples that plausibly could have been drawn from the original dataset.

Jason Brownlee

A Gentle Introduction to Generative Adversarial Networks (GANs)

<https://machinelearningmastery.com/what-are-generative-adversarial-networks-gans/>

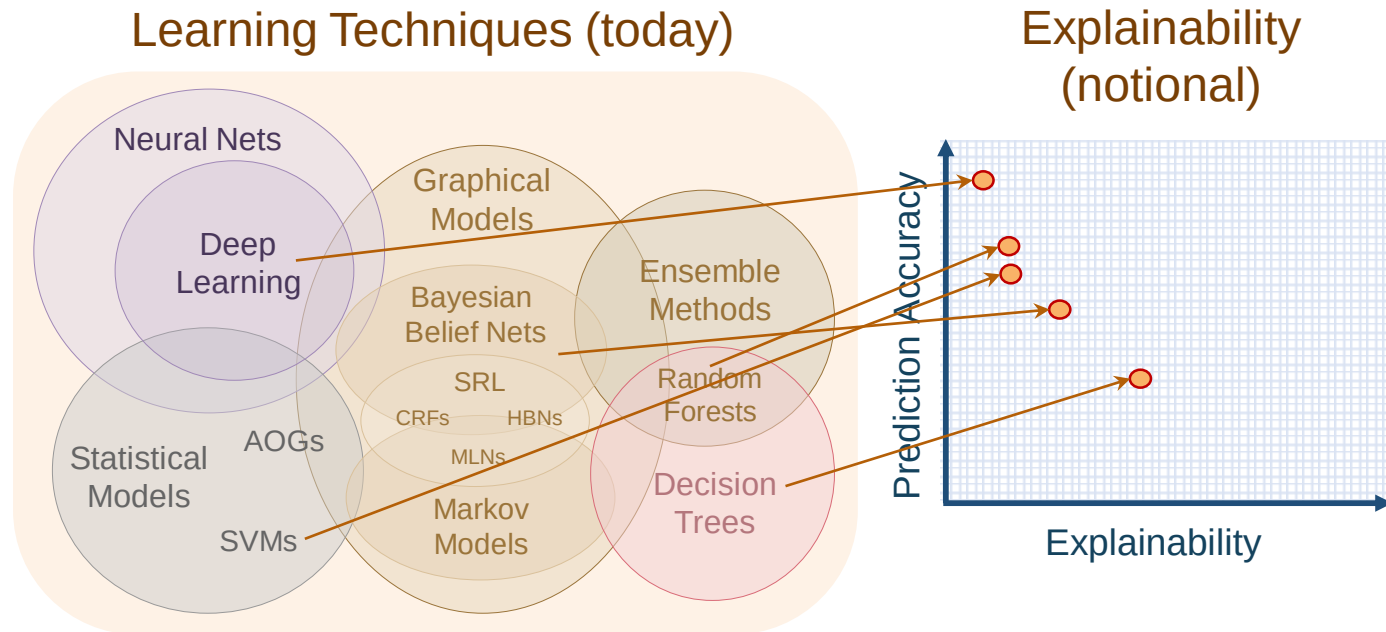
GANs Process/Architecture



Explainable AI

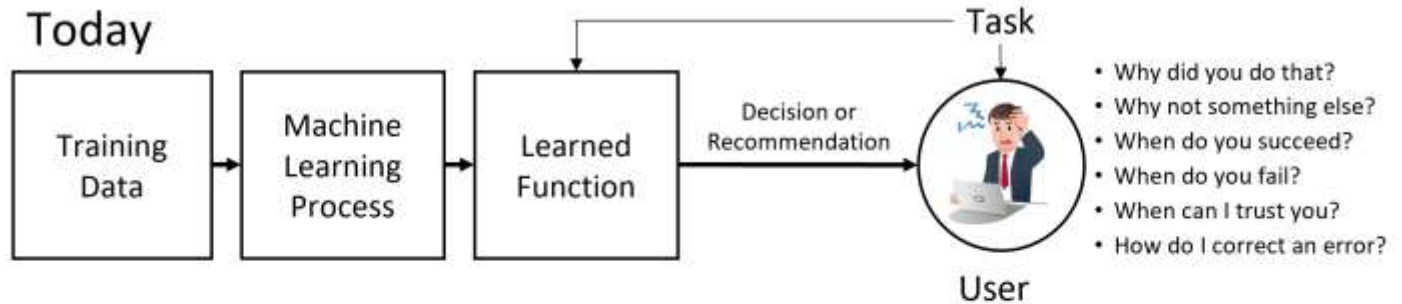
- The current generation of AI systems offer tremendous benefits, but their effectiveness will be limited by the machine's inability to explain its decisions and actions to users.
- Explainable AI will be essential if users are to understand, appropriately trust, and effectively manage this incoming generation of artificially intelligent partners.

Explainability of Today's Algorithms

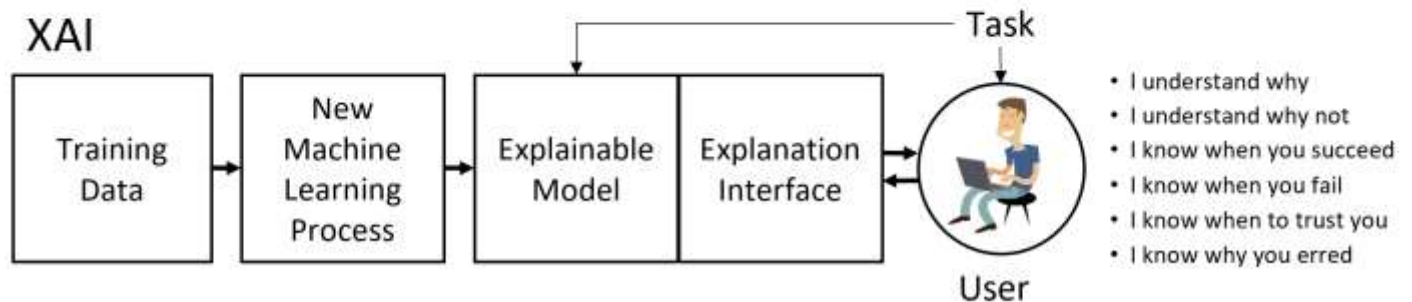


XAI vs AI

Today



XAI



Python XAI libraries

Here are 6 popular Python frameworks for model interpretability and explainability

[SHAP](#)

[LIME](#)

[Shapash](#)

[ELI5](#)

[InterpretML](#)

[OmniXAI](#) (Integrated framework of libraries above)

SHAP example

```
[1]: import numpy as np
import xgboost
from sklearn.linear_model import LinearRegression

import shap
```





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LIME

```
# Import the LimeTabularExplainer module
from lime.lime_tabular import LimeTabularExplainer

# Get the class names
class_names = ['Has diabetes', 'No diabetes']

# Get the feature names
feature_names = list(X_train.columns)

# Fit the Explainer on the training data set using the LimeTabularExplainer
explainer = LimeTabularExplainer(X_train.values, feature_names =
                                feature_names,
                                class_names = class_names,
                                mode = 'classification')
```

Prediction probabilities



Feature	Value
Glucose	104.00
Age	38.00
Pregnancies	13.00
SkinThickness	0.00
BloodPressure	72.00
DiabetesPedigreeFunction	0.47
BMI	31.20
Insulin	0.00

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Summary

In this lecture:

- We have learnt ANN, Deep Learning (DL) and different types of DL.
- We focused on two research areas of Computer Vision (CV) and NLP.
- Particularly, we had a detail study on CNN and LSTM (a special case of RNN)
- Ideas of transfer learning and GANs

Next:

NLP (Natural Language Processing) and practices